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The stability, electronic structure, elastic and metallic properties of manganese nitrides

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The phase stability, electronic structure, elastic and metallic properties of manganese nitrides (Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, Mn_3N_2 , and MnN) were extensively studied by first principles calculations. The negative values of cohesive energy and formation enthalpy show that these compounds are thermodynamically stable. The bonding of Manganese nitrides is the combinations of covalent and metallic bonds. In addition, a strong hybridization exists nearby the Fermi level, being characteristic of N-p and Mn-d states. The elastic properties of these nitrides (Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, and MnN) were calculated, which included bulk modulus, shear modulus, Young's modulus, Poisson's ratio, and hardness. The calculated results reveal that $\text{Mn}_2\text{N}_{0.86}$ and MnN are ductile while Mn_4N is brittle, and the hardness values of these nitrides are 24.35 GPa, 12.01 GPa, and 17.06 GPa, respectively. The MnN compound has the highest Debye temperature (632.1 K), while $\text{Mn}_2\text{N}_{0.86}$ has the lowest Debye temperature (390.5 K). The anisotropy of Young's modulus for Mn_4N and MnN compounds is more prominent than $\text{Mn}_2\text{N}_{0.86}$ at the three crystalline planes. Moreover, the compound of Mn_3N_2 has an unstable structure in mechanical stability.

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1 Introduction

Transition metal nitrides have recently attracted much attention due to their excellent structural, optical, electronic, and magnetic properties that can be applied to many fields, including optical and wear-resistant coatings, magnetic recording and sensing, and spintronics.^{1–3} It is well known that there are four stable intermediate phases in the Mn–N binary system, including the face-centered cubic structure Mn_4N , hexagonal $\text{Mn}_2\text{N}_{0.86}$, and tetragonal structure Mn_3N_2 and MnN . Suzuki *et al.*⁴ investigated the crystal structure and magnetic properties of the compound MnN , which was prepared as a single phase by the DC reactive sputtering method. Its crystal structure was determined to be face-centered tetragonal with the NaCl type by X-ray diffraction measurements. This compound exhibits antiferromagnetism. Yang *et al.*⁵ studied the atomic-scale structure of η -phase Mn_3N_2 (010) using scanning tunneling microscopy and first principles calculation. Ching *et al.*⁶ published the magnetic properties and structure of Mn_4N films on glass substrates. Mn_4N films with an ordered structure and (002) texture can be formed at low substrate temperatures using DC reactive magnetron sputtering from a sintered Mn target onto a glass substrate. They observed that perpendicular magnetic anisotropy exists in these films. Ciftci

*et al.*⁷ studied the structural, electronic, elastic, thermodynamic, and vibrational properties of TbN using first principles calculations. They used the projected augmented plane-wave (PAW) implementation of the Vienna *Ab initio* Simulation Package (VASP), and they obtained key results on basic thermodynamical quantities. Brik *et al.*⁸ performed first principles studies on the electronic and elastic properties of metal nitrides XN ($\text{X} = \text{Sc}, \text{Ti}, \text{V}, \text{Cr}, \text{Zr}, \text{and Nb}$) based on density functional theory (DFT).

In a previous work, we studied the stability and elastic properties of NaCl-type MN ($\text{M} = \text{Ti}, \text{V}, \text{Zr}, \text{Nb}, \text{and Ta}$)⁹ and trigonal-type M_2N ($\text{M} = \text{Cr}, \text{V}, \text{Nb}, \text{and Ta}$)¹⁰ compounds that were successfully investigated by first principles calculations. These results revealed that these transition metal nitrides in steels were stable compounds. Although manganese nitrides are common in the second phase in steel, there are few studies that have described the properties of manganese nitrides and their related properties. In the present work, we report systematic calculations for the crystal structures, stability, electronic, elastic, and metallic properties and synthesis routes of manganese nitrides (Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, Mn_3N_2 , and MnN) by first principles calculations.

2 Methods and details

In this work, the initial lattice parameters from experimentally observed compositions of Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, Mn_3N_2 , and MnN were adopted for the calculation. The calculations were carried out using the Cambridge serial total energy package (CASTEP) code,¹¹ based on density functional theory. The electron

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exchange and correlation were treated using the Perdew–Burke–Ernzerhof (PBE) formulation of the generalized gradient approximation (GGA).¹² A 360 eV cutoff energy for a reciprocal-space grid was chosen for the plane wave. The Brillouin zones were sampled by $10 \times 10 \times 10$ k -point meshes according to the Monkhorst–Pack scheme for geometric optimization. The valence electrons of distinct atoms considered were including: $2s^2 2p^3$ (N), $3d^5 4s^2$ (Mn). The thermodynamically stable criteria were identified by the cohesive energy and formation enthalpy using eqn (1) and (2).

$$E_{\text{coh}}(\text{Mn}_x\text{N}_y) = \frac{E_{\text{tot}}(\text{cell}) - xE_{\text{iso}}(\text{Mn}) - yE_{\text{iso}}(\text{N})}{x + y} \quad (1)$$

$$\Delta_f H_m(\text{Mn}_x\text{N}_y) = \frac{E_{\text{tot}}(\text{Mn}_x\text{N}_y) - xE_{\text{bin}}(\text{Mn}) - yE_{\text{bin}}(\text{N})}{x + y} \quad (2)$$

Here, $E_{\text{coh}}(\text{Mn}_x\text{N}_y)$ and $\Delta_f H_m(\text{Mn}_x\text{N}_y)$ are the cohesive energy and formation enthalpy of Mn_xN_y , respectively. $E_{\text{tot}}(\text{cell})$ is the total energy of a calculated cell; $E_{\text{bin}}(\text{Mn})$ is the cohesive energy of the pure elemental Mn atom, and $E_{\text{iso}}(\text{Mn})$ shows the total energy of an isolated Mn atom.

For the hexagonal system, the Voigt and Reuss methods were used to evaluate B and G :¹³

$$\begin{aligned} B_V &= \frac{1}{9}[2(C_{11} + C_{12}) + C_{33} + 4C_{13}] \\ G_V &= \frac{1}{30}(7C_{11} - 5C_{12} + 12C_{44} + 2C_{33} - 4C_{13}) \\ B_R &= \frac{(C_{11} + C_{12})C_{33} - 2C_{13}^2}{C_{11} + C_{12} + 2C_{33} - 4C_{13}} \\ G_R &= \frac{5}{2} \left\{ \frac{[(C_{11} + C_{12})C_{33} - 2C_{13}^2]C_{44}C_{66}}{3B_V C_{44}C_{66} + [(C_{11} + C_{12})C_{33} - 2C_{13}^2](C_{44} + C_{66})} \right\} \end{aligned} \quad (3)$$

For tetragonal structure, B and G are:¹⁴

$$\begin{aligned} B_V &= \frac{1}{9}[2(C_{11} + C_{12}) + C_{33} + 4C_{13}] \\ G_V &= \frac{1}{30}(M + 3C_{11} - 3C_{12} + 12C_{44} + 6C_{66}) \\ B_R &= C^2/M \\ G_R &= \frac{15}{18B_V/C^2 + 6/(C_{11} - C_{12}) + 6/C_{44} + 3/C_{66}} \\ M &= C_{11} + C_{12} + 2C_{33} - 4C_{13} \\ C^2 &= (C_{11} + C_{12})C_{33} - 2C_{13}^2 \end{aligned} \quad (4)$$

The Hill method calculations are as follows:

$$B_H = \frac{1}{2}(B_V + B_R) \quad (5)$$

$$G_H = \frac{1}{2}(G_V + G_R) \quad (6)$$

E and ν is calculated by B_H and G_H as following:

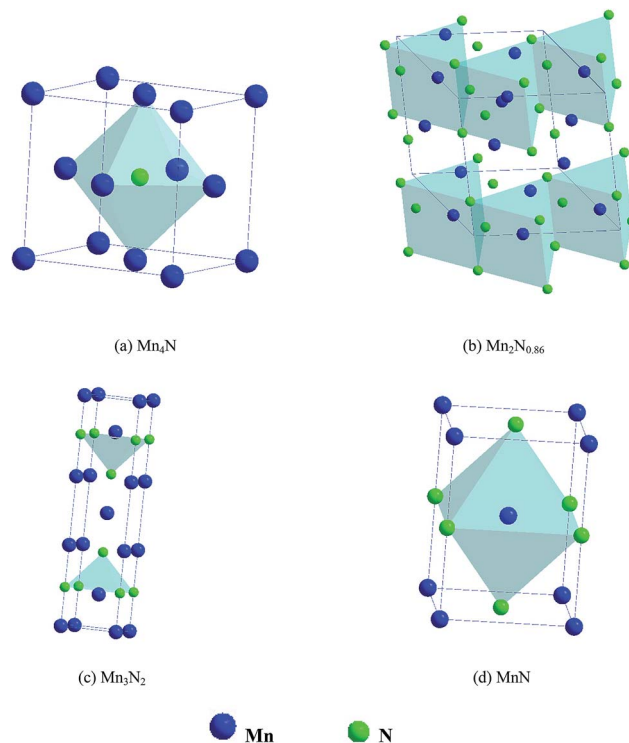


Fig. 1 (a–d) The crystal structure of manganese nitrides (Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, Mn_3N_2 , and MnN).

$$E = 9B_H G_H / (3B_H + G_H) \quad (7)$$

$$\nu = \frac{(3B_H - 2G_H)}{[2(3B_H + G_H)]} \quad (8)$$

The crystal structures of manganese nitrides are shown in Fig. 1. The cubic structure of Mn_4N contains 5 atoms, consisting of 4 Mn atoms and 1 N atom, with Wyckoff site 1a (0,0,0) and 3c (0,0.5,0.5) for Mn and 1b (0.5,0.5,0.5) for N. The hexagonal structure of $\text{Mn}_2\text{N}_{0.86}$ contains 8.58 atoms, consisting of 6 Mn atoms and 2.58 N atoms, with Wyckoff site 6g (0.3333,0,0) for Mn and 2b (0,0,0.25), 2c (0.3333,0.6667,0.25) and 2d (0.3333,0.6667,0.75) for N. For the tetragonal structures of Mn_3N_2 and MnN , Mn_3N_2 contains 10 atoms consisting of 6 Mn atoms and 4 N atoms, with Wyckoff site 2a (0,0,0) and 4e (0,0,0.3301) for Mn and 4e (0,0,0.1594) for N. MnN contains 4 atoms, consisting of 2 Mn atoms and 2 N atoms, with Wyckoff site 2a (0,0,0) for Mn and 2b (0,0,0.5) for N.

3 Results and discussion

3.1 Stability

We obtained the lattice parameters by geometric optimization as well as the cell volume, cohesive energy, and formation enthalpy of manganese nitrides; the calculated results are shown in Table 1. According to the results, the optimized geometric lattice parameters are in good agreement with the experimental values. These results imply that the methods are reliable for these compounds. Moreover, the thermodynamically

Table 1 The calculated optimized lattice parameters, cell volume, cohesive energy, and formation enthalpy of manganese nitrides

Species	Space group	No.	Experimental parameter (Å)			Optimization parameter (Å)			V (Å ³)	E_{coh} (eV per atom)	$\Delta_f H_m$ (kJ mol ^{−1})	
			a	b	c	a	b	c			Our work	Ref.
Mn ₄ N	$Pm\bar{3}m$ cubic	221	3.87 ^a	3.87 ^a	3.87 ^a	3.68	3.68	3.68	49.76	−1.800	−140.10	−127.612 ^e
Mn ₂ N _{0.86}	$P6322$ hexagonal	182	4.86 ^b	4.86 ^b	4.53 ^b	4.96	4.96	4.53	96.37	−1.796	−173.62	
Mn ₃ N ₂	$I4/mmm$ tetragonal	139	2.99 ^c	2.99 ^c	12.50 ^c	2.77	2.77	11.33	87.19	−1.758	−170.58	−191.627 ^e
MnN	$I4/mmm$ tetragonal	139	3.01 ^d	3.01 ^d	4.19 ^d	2.85	2.85	4.03	32.68	−2.152	−192.92	

^a Ref. 15. ^b Ref. 16. ^c Ref. 17. ^d Ref. 4. ^e Ref. 18.

stable crystal structure requires negative values for the cohesive energy and formation enthalpy. In the case of cohesive energy, the values for these nitrides (Mn₄N, Mn₂N_{0.86}, Mn₃N₂, and MnN) are −1.800 eV per atom, −1.796 eV per atom, −1.758 eV per atom, and −2.152 eV per atom, respectively, so that these compounds are stable phases, and the descending order of stability is Mn₃N₂ > Mn₂N_{0.86} > Mn₄N > MnN. It indicates that MnN is the most stable crystal structure, and Mn₃N₂ is the unstable phase. From the formation enthalpy, it demonstrates that MnN has a more stable phase than other nitrides.

3.2 Electronic properties

The density of states (DOS) is useful to understand the chemical bonding in a compound. To reveal the origin of the mechanical properties of these nitrides, the total and partial densities of states of manganese nitrides are calculated at the absolute temperature, as presented in Fig. 2. It can be seen from Fig. 2 that Mn₄N, Mn₂N_{0.86}, Mn₃N₂, and MnN exhibit metallic behavior because there is no band gap at the Fermi level. In terms of Mn₄N, the dominant feature of DOS curves is the existence of the sharp peaks contributed by s and p orbitals of N, which hybridize weakly with the other orbitals below −10 eV

as shown in Fig. 2. There is a strong hybridization existing near the Fermi level that is characteristic of N-p and Mn-d states. Obviously, the similar DOS profiles have been also observed in the case of Mn₂N_{0.86}, Mn₃N₂, and MnN. In general, high density of states at the Fermi level is typically associated with instability of the structure. It was found that the DOS values in the Fermi level of Mn₄N (7.33 eV), Mn₂N_{0.86} (10.37 eV), Mn₃N₂ (18.07 eV), and MnN (6.09 eV) at the absolute temperature follow the descending order of Mn₃N₂ > Mn₂N_{0.86} > Mn₄N > MnN. Therefore, MnN is the most stable compound while Mn₃N₂ is the weakest in the Mn–N binary system. This result is consistent with the front results pertaining to thermodynamic stability.

Population analysis provides more insightful information on the chemical bonding characteristics listed in Table 2. The Mulliken method is used for the overlap population and the charge calculation. The average bond length and the mean bond population are estimated by eqn (9) and (10):¹⁹

$$\bar{L}(\text{AB}) = \frac{\sum_i L_i N_i}{\sum_i N_i} \quad (9)$$

$$\bar{n}(\text{AB}) = \frac{\sum_i n_i N_i}{\sum_i N_i} \quad (10)$$

where $\bar{L}(\text{AB})$ and $\bar{n}(\text{AB})$ indicate the average bond length and the mean bond population, respectively; N_i is the total number of i bonds in the cell, and L_i is the bond length of i type.

The nominal charges carried by distinct atoms are calculated in Table 2. For all manganese nitrides, the negative charges carried by N are approximately −0.6e, and the positive charges are carried by Mn. The strongest chemical interaction in these nitrides is attributed to the N^{2p}–Mn^{3d} bond. One can propose two possible electron transfer paths, intra- or inter-, between the N and Mn atom, where intra- refers to p–d hybridized covalent bonding between N and Mn, and inter- is induced by the metallic or weak covalent bonding among Mn atoms. The electronic conductivity of these compounds depends on the redundant electrons in Mn atoms. In the case of intra-electron transfer paths, the Mn atoms donate electrons while the N atoms accept electrons to form chemical covalent bonds. However, Mn–Mn bonds or N–N bonds have a negative overlap

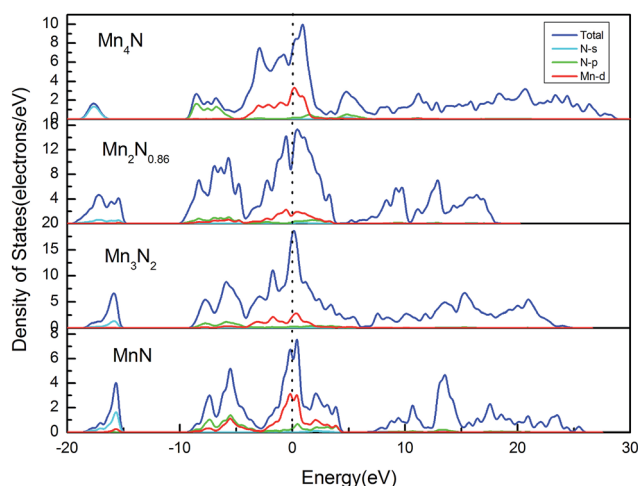


Fig. 2 Total and partial density of states of manganese nitrides from the GGA calculation. The vertical dotted line at zero indicates the Fermi energy level.

Table 2 Population analysis results of the studied nitrides, bond length (unit in Å), and the mean populations, which were calculated by Mulliken's method

Species	Atom	s	p	d	Total	Charge (e)	Bond	$\bar{L}(\text{AB})$	$\bar{n}(\text{AB})$	Net overlap electrons
Mn_4N	N	1.64	3.95	0	5.59	−0.59	N–Mn	1.839	0.35	1.40
	Mn	0.57	0.86	5.73	7.17	−0.17	Mn–Mn	2.600	−0.08	−0.48
	Mn	0.42	0.41	5.92	6.75	0.25				
$\text{Mn}_2\text{N}_{0.86}$	N	1.66	3.87	0	5.53	−0.53	N–Mn	2.003	0.22	7.92
	Mn	0.15	0.39	5.93	6.47	0.53	Mn–Mn	2.827	−0.24	−3.60
							N–N	2.663	−0.11	−0.99
Mn_3N_2	N	1.66	3.92	0	5.58	−0.58	N–Mn	1.986	0.47	5.64
	Mn	0.39	0.71	5.87	6.97	0.03	Mn–Mn	2.584	0.01	0.04
	Mn	0.26	0.33	5.85	6.44	0.56	N–N	2.688	−0.12	−0.24
MnN	N	1.68	3.92	0	5.59	−0.59	N–Mn	2.014	0.74	2.96
	Mn	0.18	0.36	5.87	6.41	0.59	Mn–Mn	2.849	−0.22	−0.22
							N–N	2.849	−0.18	−0.18

population, which illustrates anti-bond states or strong electrostatic repulsion among manganese atoms or nitride atoms. Thus, we conclude that the bonding behaviors of these nitrides are combinations of covalent and metallic bonds. Moreover, the valence bands were determined to be the contribution of both p orbitals of the nitride atom and d orbitals of the manganese atom, which is consistent with the DOS curves.

3.3 Elastic properties

The elastic stiffness constant (C_{ij}) of solids provides a link between the mechanical and dynamical behavior of crystals, and supplies important information concerning the nature of the forces operating on solids. As a qualitative check, we assessed the mechanical stability conditions for Mn–N compounds in Table 3 by diagonalizing the C_{ij} matrix and evaluating the Born stability criteria:¹⁹ for cubic systems, $(C_{11} - C_{12}) > 0$, $C_{11} > 0$, $C_{44} > 0$, $(C_{11} + 2C_{12})/3 > 0$, and $C_{12} < B < C_{11}$; for hexagonal crystals, $C_{44} > 0$, $C_{11} > C_{12}$, $(C_{11} + 2C_{12})C_{33} > 2C_{13}^2$; for tetragonal phase, $C_{11} > 0$, $C_{33} > 0$, $C_{44} > 0$, $C_{66} > 0$, $(C_{11} - C_{12}) > 0$, $(C_{11} + C_{33} - 2C_{12}) > 0$, and $2(C_{11} + C_{12}) + C_{33} + 4C_{13} > 0$. Our results for elastic constants satisfy these traditional mechanical stability criteria for all considered phases except Mn_3N_2 . Because the elastic constants of Mn_3N_2 do not satisfy traditional mechanical stability criteria, the mechanical prospects for Mn_3N_2 show that it has an unstable structure. Therefore, the mechanical properties of Mn_3N_2 are not considered.

The calculated results are shown in Table 4. The bulk modulus is resistant to shape change by applied pressure,

and the shear modulus is resistant to deformations upon shear stress. The larger the shear modulus, the more formed directional bonding there is between atoms. The calculated results demonstrate that directional bonding of Mn_4N between atoms is quite notable. Furthermore, Young's modulus provides a gauge of stiffness of the solid, whereby the larger the Young's modulus, the greater the stiffness of materials. Table 3 shows that Mn_4N has the largest bond strength among Mn–N systems.

The ratio of the bulk modulus to shear modulus of crystalline phases can predict the brittle and ductile behavior of materials. Pugh proposed that the B/G ratio should represent a measure of the 'machinable behavior'. The critical value that separates ductile and brittle materials is approximately 1.75.²⁰ When the B/G ratio is higher than 1.75, the material behaves in a ductile manner, and when lower, the material behaves in a brittle manner. The calculations reveal that Mn_4N is brittle and $\text{Mn}_2\text{N}_{0.86}$ and MnN are ductile. The Poisson's ratio can reflect the stability of a crystal against shear stress. In the present case, the ν values are 0.16 (Mn_4N), 0.36 ($\text{Mn}_2\text{N}_{0.86}$), and 0.27 (MnN).

As well known, the hardness is related to the elastic and plastic properties of the material. Vickers hardness of manganese nitrides is estimated by an empirical model²¹ based on the squared Pugh's modulus ratio ($k = G/B$) and the shear modulus (G): $H_v = 2.0(k^2G)^{0.585} - 3.0$ (H_v in GPa). It can be seen from Table 3 that the hardness values of the nitrides Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, and MnN are 24.35 GPa, 12.01 GPa, and 17.06 GPa, respectively. Our results indicate that these compounds are hard materials ($H_v > 10$ GPa), and the hardness of Mn_4N is larger than that of the other two nitrides.

Table 3 Calculated whole sets of elastic stiffness constants C_{ij} (GPa) for Mn–N compounds

Phase	C_{11}	C_{12}	C_{13}	C_{33}	C_{44}	C_{66}
Mn_4N	717.24	132.33			156.31	
$\text{Mn}_2\text{N}_{0.86}$	483.89	222.68	330.42	537.22	107.07	127.60
Mn_3N_2	307.60	495.90	188.71	545.88	94.35	253.10
MnN	536.65	291.02	105.97	630.27	122.17	247.23

3.4 Anisotropy of acoustic velocities

The sound velocities of the transverse and longitudinal modes of the manganese nitrides are calculated using previously published single crystal elastic constants.²² The sound velocities are determined by the symmetry of the crystal and the propagation direction. For example, pure transverse and longitudinal modes are found in $[001]$, $[100]$, and $[110]$ directions, and the

Table 4 The bulk modulus, shear modulus, Young's modulus E , Poisson ratio, and Vicker's hardness that were calculated using the elastic constants, with the Voigt–Reuss–Hill approximations applied for the evaluation of mechanical moduli

Species	B			G			E	B/G	ν	H_v
	B_V	B_R	B_H	G_V	G_R	G_H				
Mn_4N	327.3	327.3	327.3	210.8	210.8	210.8	676.0	1.55	0.16	24.35
$Mn_2N_{0.86}$	364.9	353.4	359.2	109.4	104.9	107.2	292.5	3.35	0.36	12.01
MnN	301.1	299.9	300.5	178.3	155.4	166.9	422.3	1.80	0.27	17.06

sound-propagating modes in other directions are quasi-transverse or quasi-longitudinal waves. In the principle directions, the acoustic velocities are simply written as:^{13,23}

Cubic (Mn_4N)

$$\begin{aligned} \text{For } [100]v_l &= \sqrt{c_{11}/\rho}; [010]v_{t1} = [001]v_{t2} = \sqrt{c_{44}/\rho} \\ \text{For } [110]v_l &= \sqrt{(c_{11} + c_{12} + 2c_{44})/2\rho}; [1\bar{1}0]v_{t1} = \sqrt{(c_{11} - c_{12})/\rho}; [001]v_{t2} = \sqrt{c_{44}/\rho} \\ \text{For } [111]v_l &= \sqrt{(c_{11} + 2c_{12} + 4c_{44})/3\rho}; [11\bar{2}]v_{t1} = v_{t2} = \sqrt{(c_{11} - c_{12} + c_{44})/3\rho} \end{aligned} \quad (11)$$

Hexagonal ($Mn_2N_{0.86}$)

$$\begin{aligned} \text{For } [100]v_l &= \sqrt{(c_{11} - c_{12})/2\rho}; [010]v_{t1} = \sqrt{c_{44}/2\rho}; [010]v_{t2} = \sqrt{c_{44}/2\rho} \\ \text{For } [001]v_l &= \sqrt{c_{33}/2\rho}; [100]v_{t1} = \sqrt{c_{44}/2\rho}; [010]v_{t2} = \sqrt{c_{44}/2\rho} \end{aligned} \quad (12)$$

Tetragonal (MnN)

$$\begin{aligned} \text{For } [100] &= [010]v_l = \sqrt{(c_{11} - c_{12})/2\rho}; [001]v_{t1} = \sqrt{c_{44}/\rho}; [010]v_{t2} = \sqrt{c_{66}/\rho} \\ \text{For } [001]v_l &= \sqrt{c_{33}/\rho}; [100]v_{t1} = [010]v_{t2} = \sqrt{c_{66}/2\rho} \\ \text{For } [110]v_l &= \sqrt{(c_{11} + c_{12} + 2c_{66})/2\rho}; [001]v_{t1} = \sqrt{c_{44}/\rho}; [1\bar{1}0]v_{t2} = \sqrt{(c_{11} - c_{12})/2\rho} \end{aligned} \quad (13)$$

With the computed transverse and longitudinal acoustic velocities, it is straightforward to calculate the Debye temperature using the equation:²⁴

$$\Theta_D = \frac{h}{k} \left[\frac{3}{4\pi} \left(\frac{N_A \rho}{M} \right) \right]^{1/3} v_m \quad (14)$$

where h is Planck's constant, k is Boltzmann's constant, N_A is Avogadro's number, n is the number of atoms in a unit cell, ρ is the density of the material, M is the mass of a unit cell, and v_m is the averaged elastic wave velocity defined as:

$$v_m = \left[\frac{1}{3} \left(\frac{2}{v_t^3} + \frac{1}{v_l^3} \right) \right]^{-1/3} \quad (15)$$

$$v_l = \sqrt{\frac{B + (4/3)G}{\rho}} \quad (16)$$

$$v_t = \sqrt{\frac{G}{\rho}} \quad (17)$$

The Debye temperature is the temperature of the highest normal mode of vibration in a crystal, and it provides insight into the thermodynamics of the material from the elastic properties. The calculated values of the Debye temperature, and the longitudinal, transverse, and average sound velocities of compounds are tabulated in Table 5. It is known that Debye temperature can be used to characterize the strength of covalent

Table 5 The anisotropic sound velocities and average velocities of manganese nitrides. The unit of velocity is km s^{-1} , and the Debye temperature (K) is also shown below

Propagation direction	[100]			[110]			[001]			v_l	v_s	v_m	Θ_D
	$[100]v_l$	$[010]v_{t1}$	$[001]v_{t2}$	$[110]v_l$	$[1\bar{1}0]v_{t1}$	$[001]v_{t2}$	$[001]v_l$	$[100]v_{t1}$	$[010]v_{t2}$				
Mn_4N	9.589	4.477	4.477							7.888	5.198	5.691	462.7
$\text{Mn}_2\text{N}_{0.86}$	7.691	8.764	4.123				9.234	4.123	4.123	4.124	8.352	4.630	390.5
MnN	8.753	5.941	4.176	9.714	4.187	4.176	9.485	5.941	5.941	4.881	7.795	5.367	632.1

bonds in solids. According to Table 5, among these nitrides, MnN has the highest Debye temperature (632.1 K), while $\text{Mn}_2\text{N}_{0.86}$ has the lowest Debye temperature (390.5 K). Therefore, MnN possesses stronger covalent bonds than other projected nitrides.

3.5 Angular variation in Young's modulus

Although the calculations of elastic constants and sound velocities indicate that the elastic properties of manganese nitrides are anisotropic, it is a useful approach to examining elastic anisotropy to use mathematically-derived surfaces that change their shape from one crystallographic direction to the next. The extent to which the surfaces change shape is indicative of anisotropy.²⁵

For cubic structures, the following equations are also suitable by assuming $S_{11} = S_{33}$, $S_{12} = S_{13}$, and $S_{44} = S_{55} = S_{66}$,

$$\frac{1}{E} = l_1^4 S_{11} + l_2^4 S_{22} + l_3^4 S_{33} + 2l_1^2 l_2^2 S_{12} + 2l_1^2 l_3^2 S_{13} + 2l_2^2 l_3^2 S_{23} + l_1^2 l_2^2 S_{66} + l_1^2 l_3^2 S_{55} + l_2^2 l_3^2 S_{44} \quad (18)$$

The directional dependence of the shear modulus for hexagonal crystals can be calculated using the following relationships:

$$\frac{1}{E} = (1 - l_3^2)^2 S_{11} + l_3^4 S_{33} + l_3^2 (1 - l_3^2) (2S_{13} + S_{44}) \quad (19)$$

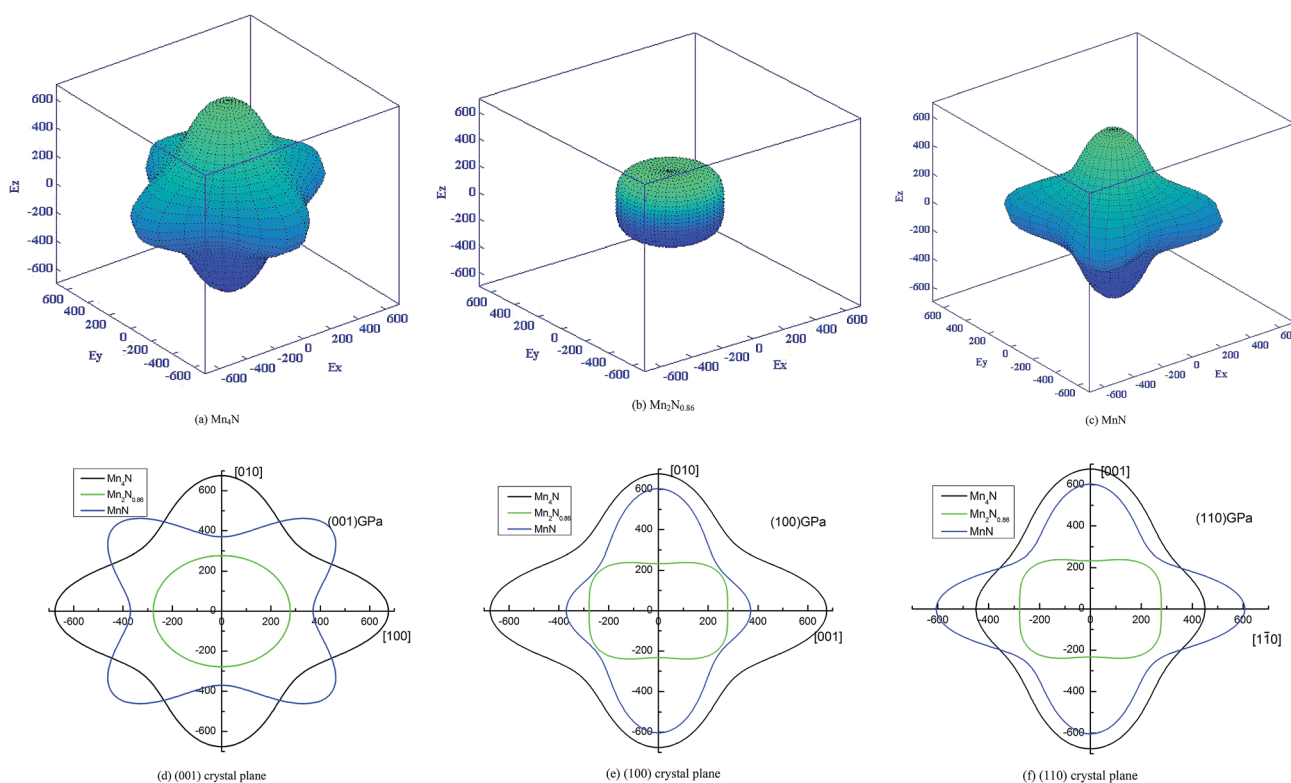


Fig. 3 (a–c) The directional dependence of the Young's modulus for manganese nitrides (Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, and MnN). (d–f) The (001), (100), and (110) planar projections, respectively, of the Young's modulus for manganese nitrides.

For tetragonal crystal structure,²⁶ the elastic constants are related to the directional cosines, l_i ,

$$\frac{1}{E} = S_{11} - l_3^4(S_{11} - S_{33}) - l_3^2(l_1^2 + l_2^2)(2S_{11} - 2S_{13} - S_{44}) - l_1^2 l_2^2(2S_{11} - 2S_{12} - S_{66}) \quad (20)$$

where the S_{ij} are the components of the elastic compliance matrix, *i.e.*, the inverse of the elastic stiffness matrix, and l_1 , l_2 , and l_3 are the components of the unit vector pointing in the crystallographic direction.

A contour plot of Young's modulus for manganese nitrides is illustrated in Fig. 3. It demonstrates that these manganese nitrides exhibit a strong anisotropic character in Young's modulus except for $\text{Mn}_2\text{N}_{0.86}$. The surface construction of Mn_4N and MnN is close to a flower shape while the surface contour of the $\text{Mn}_2\text{N}_{0.86}$ is close to a spherical shape, indicating a higher symmetry of structure. The projections on the (001), (100), and (110) planes show more details regarding the anisotropic properties of the Young's modulus. The (001), (100), and (110) planar projections of Young's modulus for the manganese nitride compounds Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, and MnN , respectively, are shown in Fig. 3(d–f). At the (001), (100), and (110) plane, the anisotropy in Young's modulus for Mn_4N and MnN is more prominent than that of $\text{Mn}_2\text{N}_{0.86}$. At the (001) plane, the values in the [010] and [100] directions of Young's modulus for Mn_4N reach the maximum values. On the contrary, the values of MnN are the minimum values. The cubic Mn_4N has a large C_{11} , which explains the strong anisotropic feature of the projected Young's modulus at the (001) and (110) plane. In addition, the anisotropic character in Young's modulus for Mn_4N at the (100) plane is the same as that of the (001) plane because of symmetry. $\text{Mn}_2\text{N}_{0.86}$ shows the weaker anisotropic feature of the projected Young's modulus at the (100) and (110) plane. In the case of the MnN phase, C_{33} is larger than C_{11} or C_{44} , which demonstrates that the phases are highly distorted along the [001] direction.

4 Conclusions

Based on the DFT calculation, the phase stability, electronic structure, elastic and metallic properties of manganese nitrides were studied. The thermodynamic stability was identified by the calculated cohesive energy and formation enthalpy, which were -2.152 eV per atom and -192.92 kJ mol⁻¹, respectively, and therefore, the MnN compound is the most stable phase. The electronic structure revealed that all projected nitrides are metallic and they have a common hybridization between N-p and Mn-d orbitals that forms the covalent bonding between Mn and N atoms. The elastic properties of the nitrides Mn_4N , $\text{Mn}_2\text{N}_{0.86}$, and MnN were calculated, and the results reveal that $\text{Mn}_2\text{N}_{0.86}$ and MnN are ductile while Mn_4N is brittle; the hardness values of these nitrides are 24.35 GPa, 12.01 GPa, and 17.06 GPa, respectively. The MnN compound has the highest Debye temperature (632.1 K), while $\text{Mn}_2\text{N}_{0.86}$ has the lowest Debye temperature (390.5 K). The anisotropy of Young's modulus for Mn_4N and MnN is more prominent than that of

$\text{Mn}_2\text{N}_{0.86}$ at the three crystal planes. The absence of soft modes in the phonon dispersion curve reflects the dynamical stability of manganese nitrides. It is hoped that our results will undergo further experimental and theoretical testing in the future.

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